# PAPER REVIEW TASK 1

## Literature Review and Synthesis: AI in Healthcare

## Review of Advancements and Challenges of Federated Learning in Medical Imaging: A Systematic Literature Review


| Submitted by, | Submitted to, |
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# Abstract

The reviewed paper examines how federated learning can support collaborative cancer diagnosis without transferring raw patient images between hospitals. Its central argument is that conventional deep learning systems need large and diverse datasets, but centralized data collection is restricted by privacy laws, institutional ownership, ethics, and security risks. Federated learning addresses this barrier by training local models at participating institutions and sharing model updates for aggregation rather than sharing the original data.

The review covers five cancer domains such as skin, lung, colorectal, brain, and breast which synthesizes datasets, preprocessing practices, model architectures, federated learning configurations, evaluation metrics, privacy methods, and clinical-deployment concerns. Its main contribution is a seven-axis taxonomy that connects technical design with clinical readiness. It also identifies persistent barriers: non-IID data, domain shift, privacy leakage from gradients, class imbalance, scarce annotation, aggregation instability, communication and computation overhead, segmentation inconsistency, and weak integration with hospital workflows.

# Problem Statement

Cancer diagnosis increasingly relies on medical images such as MRI, CT, mammography, dermoscopy, and histopathology. Deep learning models can improve classification, detection, and segmentation, but their reliability usually depends on large multi-institutional datasets. In practice, patient images remain fragmented across hospitals and cannot easily be pooled because of privacy rules, legal constraints, ethical obligations, cybersecurity risks, and institutional data ownership.

The paper argues that federated learning provides a promising answer. Each hospital trains a local model using its own data. Only updates such as parameters or gradients are sent to an aggregation process, which produces a global model and redistributes it to the institutions. This reduces the need for raw-data exchange, but it does not automatically eliminate privacy attacks, model poisoning, data heterogeneity, or operational complexity.

# Review Methodology

The authors describe a systematic literature review based on Kitchenham-style review procedures and PRISMA 2020 reporting. The search covered major databases including IEEE Xplore, PubMed, SpringerLink, ScienceDirect, Scopus, ACM Digital Library, Google Scholar, and arXiv. The search combined federated-learning terms with cancer types and imaging tasks such as classification, detection, and segmentation. The screening process moved from 537 identified records to 123 included studies.

The paper’s research questions examine application areas, commonly used architectures, datasets and preprocessing, performance relative to centralized or local training, privacy risks, security mechanisms, practical hospital deployment, scalability, and interpretability. This broad question set supports the assignment because it directly covers technical trends, security challenges, open research gaps, and clinical relevance.

# Taxonomy of Federated Learning in Cancer Imaging

The paper organizes the field along seven orthogonal axes. This taxonomy is important because it prevents a narrow comparison based only on model accuracy and instead asks whether a method is secure, reproducible, interpretable, and clinically usable.

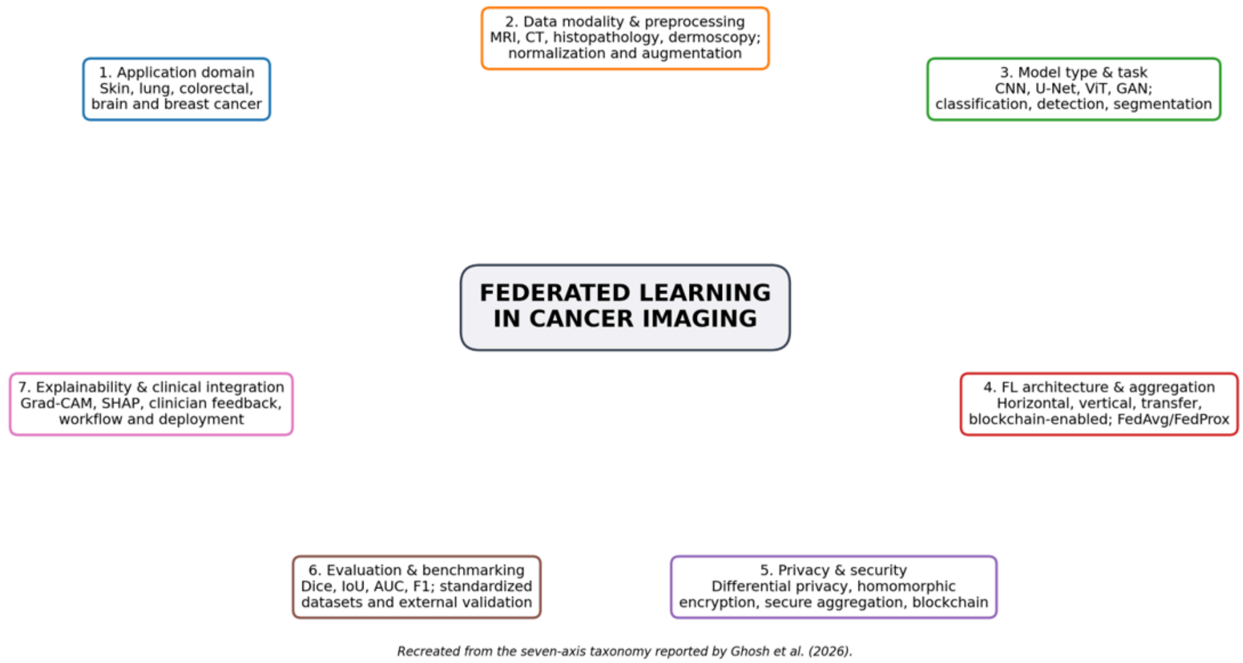


Figure 1. Seven-axis taxonomy recreated from Ghosh et al. (2026).

# Technical Synthesis and Methodological Trends

## Applications Across Five Cancer Domains

Skin cancer: The review emphasizes dermoscopic datasets such as ISIC-2019 and HAM10000. Common approaches include asynchronous FL, CNN-SVM hybrids, InceptionV3, transfer learning, and fairness-aware personalization.

Lung cancer: Studies use CT, chest X-ray, and histopathology datasets. Models include CapsNet, DenseNet, attention-based CNNs, U-Net variants, GAN-based systems, and lightweight CNNs.

Colorectal cancer: Histopathology dominates this domain. The review discusses ResNeXt, VGG/BiT, federated style transfer, FedDropoutAvg, and encrypted FL.

Brain cancer: MRI-based tumor classification and segmentation commonly use U-Net, 3D U-Net, VGG16, DenseNet, and personalized FL. BraTS is the most visible benchmark..

Breast cancer: Mammography, ultrasound, pathology images, and tabular records are used with YOLO, ResNet, U-Net, DCNN, MobileNet, autoencoders, and differential privacy.

## Repeated Architectural and Experimental Trends

1. CNN families remain dominant because they are mature and effective for image classification.
2. U-Net and 3D U-Net are central to tumor segmentation, especially for MRI and CT.
3. ResNet and DenseNet are frequently used as feature extractors because skip or dense connections support deeper learning.
4. MobileNet and EfficientNet are attractive when edge devices or smaller hospitals have limited computing resources.
5. FedAvg is still the common baseline, while FedProx, FedBN, SCAFFOLD, clustered personalization, asynchronous updates, and knowledge distillation are explored for non-IID data and communication efficiency.
6. Preprocessing is highly modality-dependent: MRI uses skull stripping and intensity normalization; CT uses denoising and lung/nodule segmentation; histopathology uses stain normalization and tile selection; dermoscopy relies on resizing, normalization, and augmentation.
7. Accuracy alone is insufficient. Segmentation requires Dice/IoU, imbalanced classification needs sensitivity/F1/AUC, and deployment studies should also report communication, latency, energy, privacy budget, and external validation.

## Security and Privacy Analysis

Federated learning improves privacy by keeping raw images at the local institution, but the review clearly shows that this is not equivalent to complete confidentiality. Shared gradients or parameters can reveal information through model-inversion and membership-inference attacks. Malicious clients may also poison local training data or upload manipulated updates that reduce global-model reliability.

Best-practice synthesis from the review: a practical medical FL system should combine secure aggregation with a lightweight encryption or differential-privacy layer, authenticate clients, log model provenance, detect anomalous updates, and report the accuracy–privacy–latency trade-off. No single technique solves every risk.

# Open Challenges Identified by the Authors

Data heterogeneity and domain shift: Hospitals differ in scanners, imaging protocols, staining, demographics, disease prevalence, and annotation. These non-IID distributions cause client drift, unstable convergence, and reduced external generalization.

Privacy and security trade-offs: Stronger DP or encryption improves protection but adds noise, computation, bandwidth, and latency. Low-resource hospitals may be unable to run heavy cryptographic protocols.

Class imbalance and annotation scarcity: Rare malignant cases are underrepresented, while expert labeling is costly and inconsistent. Federated silos can magnify the imbalance.

Aggregation instability: FedAvg may perform poorly under statistical and system heterogeneity. Advanced methods still do not fully solve divergent objectives, unreliable clients, or hardware disparities.

Computation and model optimization: 3D U-Net, YOLO, GANs, and transformer systems can be too large for edge devices or smaller hospitals. Pruning and quantization may reduce accuracy.

Segmentation robustness: Tumor boundaries and labels vary between experts and institutions. Cross-site protocol variation can reduce Dice scores and clinical reliability.

Workflow and clinical integration: Most studies do not demonstrate integration with EHR, PACS, radiology workflow, governance, regulatory review, or continuous post-deployment monitoring.

# Future Research Directions

1. Federated transfer learning for rare cancers: Use pretrained public models and local fine-tuning to improve learning when each institution has only a few rare cases.
2. Explainable federated AI: Embed Grad-CAM, SHAP, saliency, and explanation - consistency tests in local and global models to support clinician trust.
3. Multi-institutional collaboration frameworks: Develop governance, contribution rules, cross-site validation, and equitable benefit-sharing for real hospital consortia.
4. Communication-efficient FL: Use selective updates, compression, low-rank transmission, knowledge distillation, asynchronous participation, and edge–cloud coordination.
5. Federated meta-learning and personalization: Learn global initializations that rapidly adapt to each hospital’s patient population, scanners, and protocols.
6. Improved FL architectures for cancer diagnosis: Explore hierarchical, decentralized, cross-silo, and hybrid architectures designed for medical imaging rather than generic mobile FL.
7. Model optimization for deployment: Apply pruning, quantization, parameter-efficient fine-tuning, efficient backbones, and hardware-aware client selection.

# Conclusion

The reviewed paper shows that federated learning can enable collaborative cancer-imaging AI without centralizing raw patient data. Across skin, lung, colorectal, brain, and breast cancer, the literature demonstrates strong potential for classification, detection, and segmentation. The most common technical foundations are CNNs, U-Net variants, ResNet/DenseNet, EfficientNet/MobileNet, and increasingly transformers and hybrid architectures.

The main conclusion is that privacy-preserving collaboration is feasible, but real-world success depends on more than keeping data local. Secure aggregation, differential privacy, encryption, attack detection, robust personalization, communication efficiency, standardized multi-site evaluation, explainability, and clinical workflow integration are all necessary. Future research should prioritize prospective multi-institutional validation and report not only accuracy but also fairness, privacy, security, communication cost, resource use, and clinical utility.

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# One-page comparison table of selected references

Values and limitations are summarized from Table 6 of Ghosh et al. (2026); primary studies were not independently re-evaluated for this assignment.

| Study | Cancer / data | Preprocessing | FL model/task | Reported result | Privacy/security | Key limitation |
|---|---|---|---|---|---|---|
| **Yaqoob et al. (2023)** | Skin; ISIC-2019 | Augmentation | Async-FL-CNN; classification | F1 94.8%; sensitivity 94.1%; precision 96.7% | Raw data local; no extra mechanism detailed in review table | Resource heterogeneity and real-world asynchronous constraints not evaluated. |
| **Alasbali et al. (2025)** | Skin; ISIC/HAM10000/DermNet | Synthetic images; resize to 299×299 | InceptionV3 at edge nodes | Accuracy 98.6%; F1 98.0% | Secure aggregation | More edge nodes may cause delay; model-inversion risk and unseen-data generalization remain. |
| **Heidari et al. (2023)** | Lung; CIA, KDSB, LUNA16, local CT | Normalization, denoising, histogram equalization | Federated CapsNet; detection | Accuracy 99.69%; F1 99.10% | Blockchain-enabled FL | Depends on diverse, high-quality CT and sufficient annotation; deployment overhead. |
| **Saha et al. (2024)** | Lung; Kaggle CT | CLAHE, normalization, white balance, resize | Lung-AttNet (CNN+LGAM) | Accuracy 92.0%; AUC 98.43% | No DP/secure aggregation reported | Performance falls as clients increase; simulated setting and non-IID sensitivity. |
| **Truhn et al. (2024)** | Colorectal; Epi700, DACHS, TCGA, QUASAR, YCR-BCIP | Tile removal, Macenko stain method, undersampling | SHEFL; encrypted image analysis | Dice roughly 78.5%–86.8% across cohorts | Homomorphic-encryption-based secure FL | Tested within one institutional network; constant communication-cost assumption. |
| **Gunesli et al. (2023)** | Colorectal; TCGA CRC-DX | Resize, normalize, augment | FedDropoutAvg; tumor detection | F1 90% external; 84% local | No detailed privacy analysis | Possible parameter leakage/model inversion; multi-site privacy not fully examined. |
| **Kumar et al. (2024)** | Brain; BraTS 2020 MRI | Skull stripping, co-registration, labels | 3D U-Net; segmentation | Accuracy 88.60% | Blockchain and privacy-preserving FL | Blockchain adds latency/compute; extreme client heterogeneity affects consistency. |
| **Wagner et al. (2025)** | Brain; ATLAS, BRATS, MSSEG, TBI; unseen ISLES/Tumor2 | Skull stripping, 1-mm resampling, z-score | FedUniBrain; multi-disease MRI segmentation | Dice 69.1% source; 64.0% unseen | Raw data local; client adaptation | Generalization drops without client-specific adaptation; missing-modality simulation is incomplete. |
| **Lam et al. (2024)** | Breast; DDSM, CBIS-DDSM, MIAS, INBREAST, BUSI | Resize and geometric transforms | U-Net + GMM; FedBN/FedAvg segmentation | Dice 81.4% FedBN vs 80.3% FedAvg | Privacy-preserving FL setting | High-resolution images, heterogeneity, and inter-node communication complexity. |
| **Gupta et al. (2025)** | Breast; BreakHis and BUSI | Resize, normalize, client augmentation, boxes | YOLOv6 ensemble FL; classification/detection | Accuracy 98% BreakHis; 97% BUSI | Raw data local; no extra mechanism detailed | Only five simulated clients; no latency, hardware heterogeneity, or full hyperparameter reporting. |